

Sowhardo Bin Atik

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GitHub | Portfolio | LinkedIn

Objective

Backend-focused Full-Stack Developer passionate about scalable systems, backend architecture, and understanding how real-world software works internally. Building practical AI-powered features that solve real use cases and improve performance.

Education

BRAC University, Bangladesh

2022 – Jun 2026 (Expected)

Bachelor of Science in Computer Science and Engineering

CGPA: 3.69/4.00

Status: Coursework completed ; thesis defense scheduled for June 2026

Thesis: Soft Tissue Sarcoma (STS) Subtype Classification using Custom Deep Learning Models and Multi-Omics Data Integration

Technical Skills

Languages	JavaScript, Python, Java (OOP), C, C++
Backend	Node.js, Express.js, FastAPI, REST APIs, Authentication, Redis, Nginx
Frontend	React.js, Tailwind CSS
Databases	PostgreSQL, MongoDB, Indexing, ACID, Query Optimization, Vector Database
DevOps / Tools	Docker, Linux, Bash, Git, SSH, SCP, Vim, Load Testing (Autocannon), Virtual Machine
Networking	HTTP/HTTPS, SSL/TLS, DNS, TCP/IP, CDN, Reverse Proxy
CS Fundamentals	Operating Systems, Computer Networks, OOP, Computer Architecture, Cloud Computing
Problem Solving	LeetCode [Profile]

Projects

MetroSheba: AI-Driven Transit & Ticketing Platform (Express.js, PostgreSQL, React) [Live] | [GitHub]

- Developed a full-stack system with JWT-based dual-token auth (Access/Refresh) and HttpOnly cookie rotation for secure cross-origin session management between Vercel and Render.
- Built a **RAG-based AI Assistant** using **pgvector**, **Gemini Embeddings**, and **GROK LLM** to provide context-aware transit directions via semantic search within PostgreSQL.
- Engineered a real-time ticketing engine with automated validity logic and fare calculations using optimized relational schemas and connection-pooled queries.
- Performed load testing via **Autocannon**, achieving ~798 req/sec at 100–150 concurrent users; identified DB saturation points where over-parallelization caused a 334% latency spike.
- For local development, used docker containers to run database instance.

Remote Deployment Pipeline (Express.js, Bash, OpenSSH, Linux)

[Video] | [GitHub]

- Implemented an automated CI/CD-lite pipeline using Bash and SCP to synchronize local source code with a remote Linux server, reducing deployment overhead to a single command.
- Emphasized on server security by implementing Ed25519 key-based authentication (PKI) and configured a cross-platform Bridged Network environment for secure, remote API access.

Multi-Agent Research Assistant (FastAPI, React, Groq LLM)

[GitHub]

- A modular agent orchestration system, built using FastAPI and Groq API, that implements task delegation and response aggregation logic.
- Optimized research efficiency by automating cross-domain data synthesis and multi-agent validation to deliver structured, high-accuracy insights.

Achievements

AI CERTs Prompt Engineer Level 1 Certification

2024

Outcome: Through the knowledge of LLM architecture fundamentals, I am able to apply optimized prompts using zero-shot/few-shot prompting, chain of thought techniques and Role prompting.